Gaussian Models and Simple Systems in Gaussian Noise

Oguzhan Oguz
KTH Royal Institute of Technology

***Abstract*—This report analyzes univariate and bivariate Gaussian data and simple linear systems in noise. For i.i.d. samples, the behavior of the sample mean, variance, and ECDF is examined across sample sizes. For the bivariate case, the joint density, conditionals, and linear combinations are derived and verified on datasets with distinct correlations. Periodograms are used to identify tones under white and colored noise. Finally, spectra of an AR(1) stage and its IIR cascade, and the closed-form autocorrelation of the cascade output, are obtained and validated.**

***Index Terms*—AR(1) processes, autocorrelation, bivariate normal, colored noise, empirical cumulative distribution function, Gaussian distribution, linear systems, maximum likelihood estimation, periodogram, power spectral density, spectral estimation, white noise.**

I. Task 1: Univariate Gaussian Estimation

A real-valued sequence is modeled as i.i.d. samples from a Gaussian law with unknown mean and variance. In the project data the reference model is $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 0.5$ and $\sigma^2 = 2$. The goal is to estimate (μ, σ^2) from three sequences of different lengths and to examine how the empirical distribution approaches the theoretical Gaussian CDF as N grows.

For a sample $x_1, \dots, x_N$, the estimators used in the lectures are

$$\begin{aligned} \hat{\mu} &= \frac{1}{N}\sum_{i=1}^{N} x_i, \\ \hat{\sigma}^2_{\text{unb}} &= \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \hat{\mu})^2, \\ \hat{\sigma}^2_{\text{ML}} &= \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})^2. \end{aligned} \tag{1}$$

Numerical results. Applying (1) to the three sequences yields:

Sequence	N	$\hat{\mu}$	$\hat{\sigma}^2_{\text{unb}}$ / $\hat{\sigma}^2_{\text{ML}}$
1	10	1.3829	6.2650 / 5.6385
2	100	0.6135	2.1931 / 2.1711
3	1000	0.4408	1.9615 / 1.9596

The empirical distribution is described by the empirical CDF $\hat{F}_N(x) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{x_i \le x\}$. ECDFs for the three sequences are overlaid with the theoretical $\mathcal{N}(0.5, 2)$ CDF in Fig. 1.

II. Task 2: Bivariate Gaussian Geometry

Two paired datasets $\{(x(n), y(n))\}_{n=1}^{N_i}$ are provided, each generated from a bivariate Gaussian distribution with equal marginal variances $\sigma_x^2 = \sigma_y^2 = \sigma^2$ and correlation $\rho \in \{0.25, 0.75\}$. The aim is to state the joint pdf, visualize the joint behavior with a 3-D histogram, and identify which dataset corresponds to which correlation value. Finally, the expected change for a negative correlation is described.

Under the general model $\boldsymbol{z} = [x\ y]^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_x^2 & \rho\,\sigma_x\sigma_y \\ \rho\,\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}, \quad |\rho| < 1,$$

the joint density is

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} - \frac{2\rho\,(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \frac{(y-\mu_y)^2}{\sigma_y^2}\right]\right).$$

Equation (2) implies elliptical level sets whose geometry is fixed by $\boldsymbol{\Sigma}$. With equal variances, the principal axes are the $\pm 45^\circ$ directions and their lengths scale as $\sigma\sqrt{1 \pm \rho}$. Hence increasing $|\rho|$ increases eccentricity while keeping the center at (μ_x, μ_y). The sign of ρ flips the tilt between the $y{=}x$ diagonal ($\rho > 0$) and the $y{=}-x$ anti-diagonal ($\rho < 0$). The peak height of the pdf grows like $1/\sqrt{1-\rho^2}$ because the mass becomes more concentrated along the main axis as $|\rho|$ increases.

For each dataset a 3-D histogram of (x, y) is plotted (Fig. 2). The surface serves as a plug-in estimator of f_{XY}. Its overall shape footprint and tilt do not depend on the exact vertical scaling of the bins. The sample correlations computed from the data are close to 0.25 for the first set and 0.75 for the second set. This agrees with the visual impression. Set 1 appears mildly elongated, whereas Set 2 is clearly stretched along the $y{=}x$ direction. Therefore Set 1 is matched to $\rho \approx 0.25$ and Set 2 to $\rho \approx 0.75$.

If the correlation were negative with the same magnitude, e.g., $\rho = -0.25$ or $\rho = -0.75$, the ellipse would have identical axis lengths $\sigma\sqrt{1 \pm |\rho|}$ but would be tilted along $y{=}-x$. High values of x would tend to coincide with low values of y, and vice versa. Thus the sign of ρ controls orientation while its magnitude controls eccentricity and concentration.

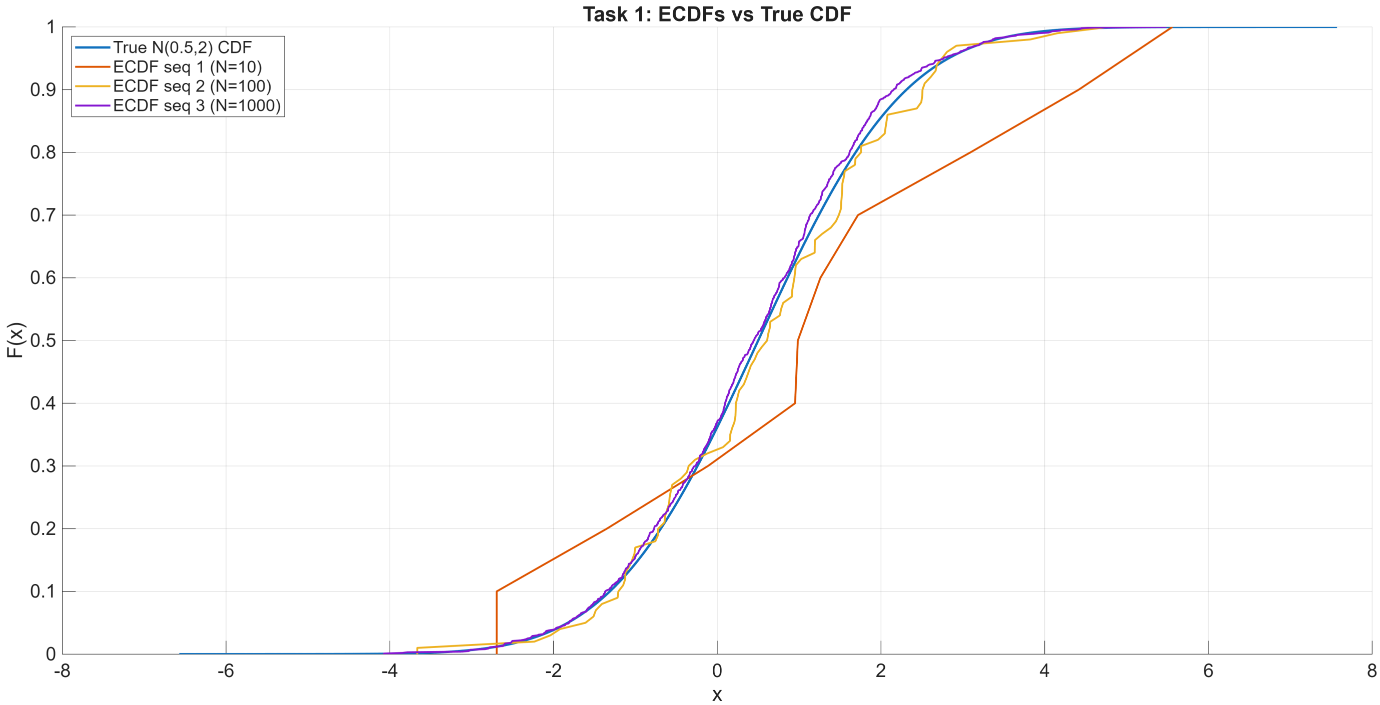
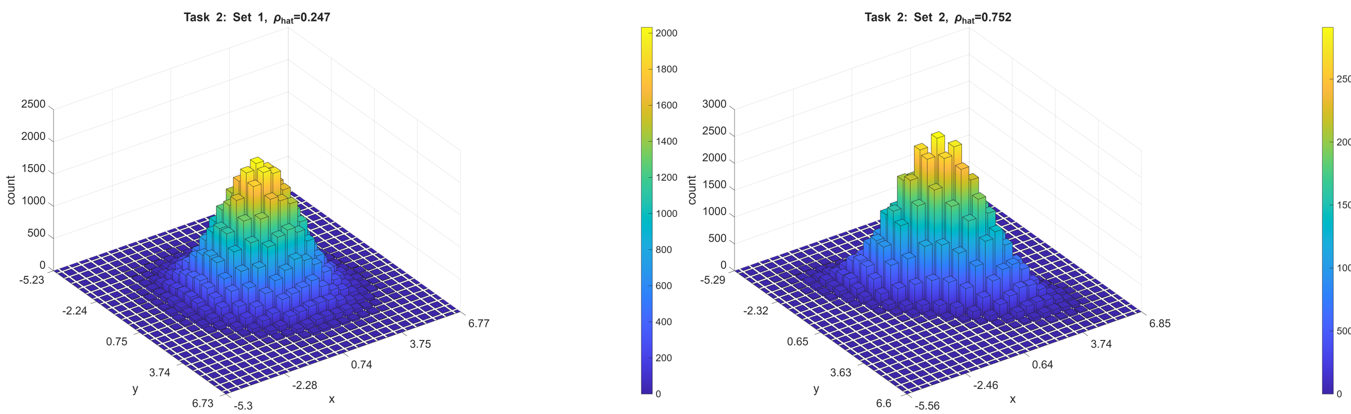


Fig. 1. ECDFs for $N \in \{10, 100, 1000\}$ overlaid with the true $\mathcal{N}(0.5, 2)$ CDF.



(a) Set 1: 3-D histogram for $\hat{\rho}=0.247$.

(b) Set 2: 3-D histogram for $\hat{\rho}=0.752$.

Fig. 2. Joint behavior of (X, Y).

III. Task 3: Conditional Laws and Linear Combinations

We now assume the equal-variance setting requested in the task: $\sigma_x^2 = \sigma_y^2 = \sigma^2$ with correlation ρ (and general means μ_x, μ_y). Starting from the exponent of the joint pdf in (2) and completing the square in x, for fixed y,

$$\begin{aligned} &\frac{(x-\mu_x)^2}{\sigma^2} - \frac{2\rho\,(x-\mu_x)(y-\mu_y)}{\sigma^2} + \frac{(y-\mu_y)^2}{\sigma^2} \\ &= \frac{\big(x-\mu_x-\rho\,(y-\mu_y)\big)^2}{\sigma^2} + (1-\rho^2)\,\frac{(y-\mu_y)^2}{\sigma^2}. \end{aligned} \tag{3}$$

The second term in (3) is constant w.r.t. x and is absorbed into the normalization of $f_{X|Y=y}$. Hence the conditional is Gaussian:

$$X \mid Y = y \sim \mathcal{N}\big(\mu_x + \rho\,(y-\mu_y),\ (1-\rho^2)\sigma^2\big), \tag{4}$$

with closed-form conditional pdf

$$f_{X|Y=y}(x) = \frac{1}{\sqrt{2\pi(1-\rho^2)}\,\sigma}\,\exp\!\left(-\frac{\big[x-\mu_x-\rho(y-\mu_y)\big]^2}{2(1-\rho^2)\sigma^2}\right). \tag{5}$$

Any linear combination of jointly Gaussian variables is Gaussian. In the equal-variance case,

$$\mu_{X+Y} = \mu_x + \mu_y, \qquad \mathrm{Var}(X+Y) = 2\sigma^2(1+\rho), \tag{6}$$

$$\mu_{X-Y} = \mu_x - \mu_y, \qquad \mathrm{Var}(X-Y) = 2\sigma^2(1-\rho). \tag{7}$$

Thus the closed-form *pdfs* are

$$f_{X+Y}(u) = \frac{1}{\sqrt{4\pi\,\sigma^2(1+\rho)}}\,\exp\!\left(-\frac{\big(u-(\mu_x+\mu_y)\big)^2}{4\sigma^2(1+\rho)}\right), \tag{8}$$

$$f_{X-Y}(v) = \frac{1}{\sqrt{4\pi\,\sigma^2(1-\rho)}}\,\exp\!\left(-\frac{\big(v-(\mu_x-\mu_y)\big)^2}{4\sigma^2(1-\rho)}\right). \tag{9}$$

IV. Task 4: Periodogram—Sinusoids in Additive White Noise

We observe a real sequence $y(n)$ that under the two hypotheses is

$$\mathcal{H}_0: \quad y(n) = w(n), \tag{10}$$

$$\mathcal{H}_1: \quad y(n) = \sum_{k=0}^{1} a_k \sin(2\pi v_k n + \phi_k) + w(n), \tag{11}$$

where $w(n)$ is zero-mean white Gaussian noise with variance σ_w^2 and $\mathrm{SNR}_k = \frac{a_k^2}{4\sigma_w^2} = 0\,\mathrm{dB}$ for each sinusoid.

For each provided sequence from `SinusInNoise1`, we compute the one-sided periodogram

$$\hat{S}_y(\nu) = \frac{1}{N}\Big|\sum_{n=0}^{N-1} y(n)\, e^{-j2\pi\nu n}\Big|^2, \qquad \nu \in [0, 0.5].$$

In Fig. 3, Seq 2 shows two isolated and very narrow lines at $\hat{v}_0 \approx 0.05$ and $\hat{v}_1 \approx 0.25$, which matches $\mathcal{H}_1$ in (11). In contrast, Seq 1 does *not* exhibit lines at 0.05 or 0.25. Its largest ordinates are around ~ 0.39 and a smaller one near ~ 0.18, with broader shapes and much lower line-to-floor contrast. This behaviour is consistent with $\mathcal{H}_0$ (white-noise only) where random fluctuations occasionally produce local maxima, rather than deterministic spectral lines.

Frequency picks are taken on the FFT grid with spacing $\Delta\nu = 1/N$, so the nominal rounding error is at most $\pm\Delta\nu/2$. Thus the recovered tones for Seq 2 are reported as $\hat{v}_0 = 0.05 \pm \Delta\nu/2$ and $\hat{v}_1 = 0.25 \pm \Delta\nu/2$.

V. Task 5: Impact of Colored Noise on Frequency Estimation

We examine `SinusInNoise2` with the periodogram and compare it to `SinusInNoise1`. The three spectra are overlaid in Fig. 4.

The colored–noise spectrum shows an elevated low–frequency floor and a broad rise towards DC, while remaining comparatively flat near mid–band. For $v_0 \approx 0.05$, the local noise floor is high, so the peak contrast is poor. The sinusoid around 0.05 becomes hard to detect and its estimated frequency is less reliable. For $v_1 \approx 0.25$: the floor is low and the spectral line is sharp and high. The estimated frequency remains accurate and stable, similar to the white–noise case.

VI. Task 6: Spectra of an AR(1) Stage and a Stable IIR Cascade

Let

$$x_1(n) = \alpha\, x_1(n-1) + z(n), \qquad \alpha = 0.25,$$

where $z(n)$ is a zero-mean white-noise sequence with variance $\sigma_z^2 = 1$. The AR(1) frequency response is $H_1(e^{j2\pi\nu}) = (1-\alpha e^{-j2\pi\nu})^{-1}$. Thus

$$\begin{aligned} S_{x_1}(\nu) &= \sigma_z^2\,\big|H_1(e^{j2\pi\nu})\big|^2 \\ &= \frac{\sigma_z^2}{1+\alpha^2-2\alpha\cos(2\pi\nu)} \\ &= \frac{1}{1.0625-0.5\cos(2\pi\nu)}. \end{aligned} \tag{12}$$

Now feed x_1 to a stable one-pole LTI with impulse response $h_2(n) = \beta^n u[n]$, $\beta = 0.25$. Since $H_2(e^{j2\pi\nu}) = (1-\beta e^{-j2\pi\nu})^{-1}$,

$$S_{x_2}(\nu) = S_{x_1}(\nu)\,\big|H_2(e^{j2\pi\nu})\big|^2 = \frac{1}{\big(1.0625-0.5\cos(2\pi\nu)\big)^2}. \tag{13}$$

Figure 5 shows the analytic PSDs.

VII. Task 7: Autocorrelation of the Cascaded Output

Convolution derivation. The overall impulse response is

$$\begin{aligned} g(n) &= (\alpha^n u[n]) * (\beta^n u[n]) = \sum_{m=-\infty}^{\infty} \alpha^m u[m]\,\beta^{\,n-m} u[n-m] \\ &= \sum_{m=0}^{n} \alpha^m \beta^{\,n-m} = \beta^{\,n} \sum_{m=0}^{n}\left(\frac{\alpha}{\beta}\right)^m = \beta^{\,n}\,\frac{1-(\alpha/\beta)^{n+1}}{1-\alpha/\beta}\,u[n] \\ &= \frac{\beta^{\,n+1}-\alpha^{\,n+1}}{\beta-\alpha}\,u[n], \qquad \alpha \neq \beta. \end{aligned} \tag{14}$$

With white input,

$$r_{x_2}(k) = \sigma_z^2 \sum_{n=-\infty}^{\infty} g(n)\,g(n+k), \qquad r_{x_2}(-k) = r_{x_2}(k). \tag{15}$$

Carrying out the geometric-series sums gives, for $k \geq 0$,

$$\boxed{r_{x_2}(k) = \frac{\sigma_z^2}{(\beta-\alpha)^2}\left[\frac{\beta^{\,k+2}}{1-\beta^2} - \frac{\alpha\,\beta^{\,k+1}}{1-\alpha\beta} - \frac{\beta\,\alpha^{\,k+1}}{1-\alpha\beta} + \frac{\alpha^{\,k+2}}{1-\alpha^2}\right].} \tag{16}$$

In the equal-pole limit $\alpha = \beta = a$,

$$\boxed{r_{x_2}(k) = \sigma_z^2\, a^{\,k}\left[\frac{k+1}{(1-a^2)^2} + \frac{2a^2}{(1-a^2)^3}\right], \qquad k \geq 0.} \tag{17}$$

Project parameters. For $a = \alpha = \beta = \frac{1}{4}$ and $\sigma_z^2 = 1$,

$$\boxed{r_{x_2}(k) = \left(\frac{1}{4}\right)^k \left[\frac{256}{225}(k+1) + \frac{512}{3375}\right], \qquad k \geq 0.} \tag{18}$$

These expressions agree with IFFT-based and empirical estimates in the Figure 6.

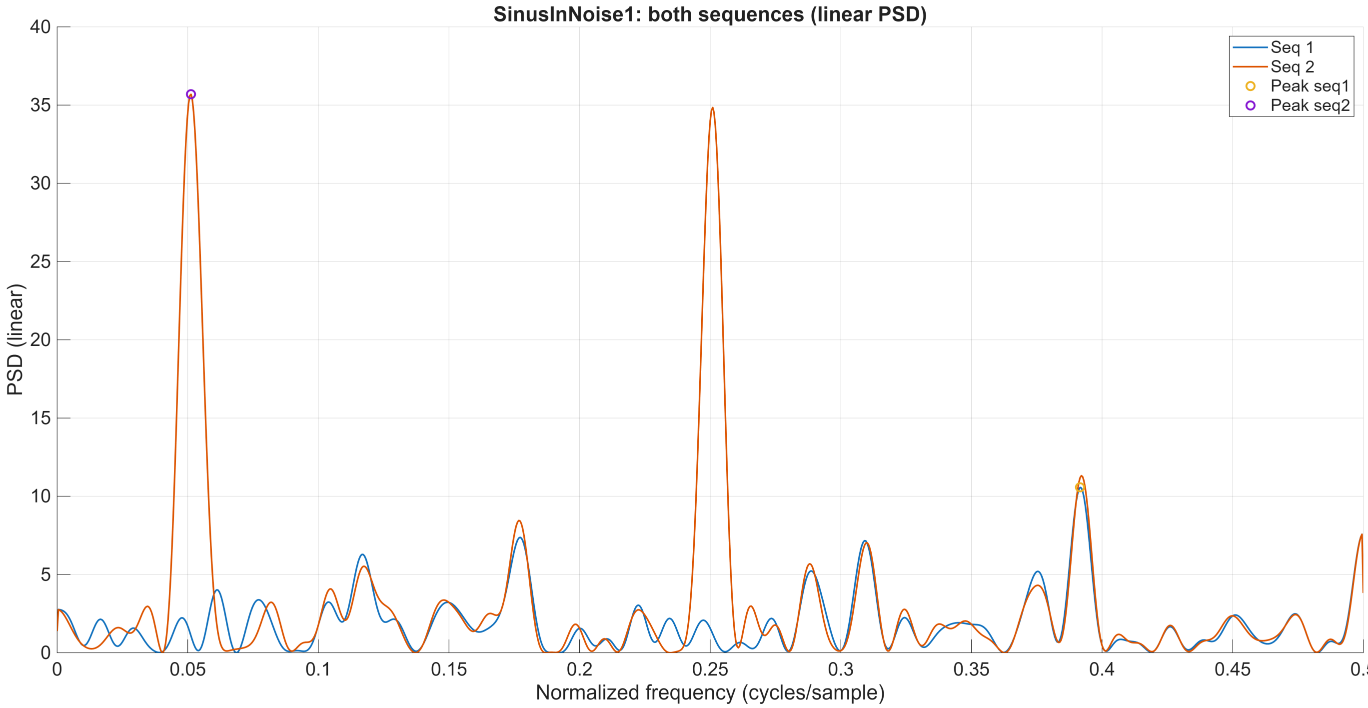


Fig. 3. Periodograms for both `SinusInNoise1` sequences on the same axes.

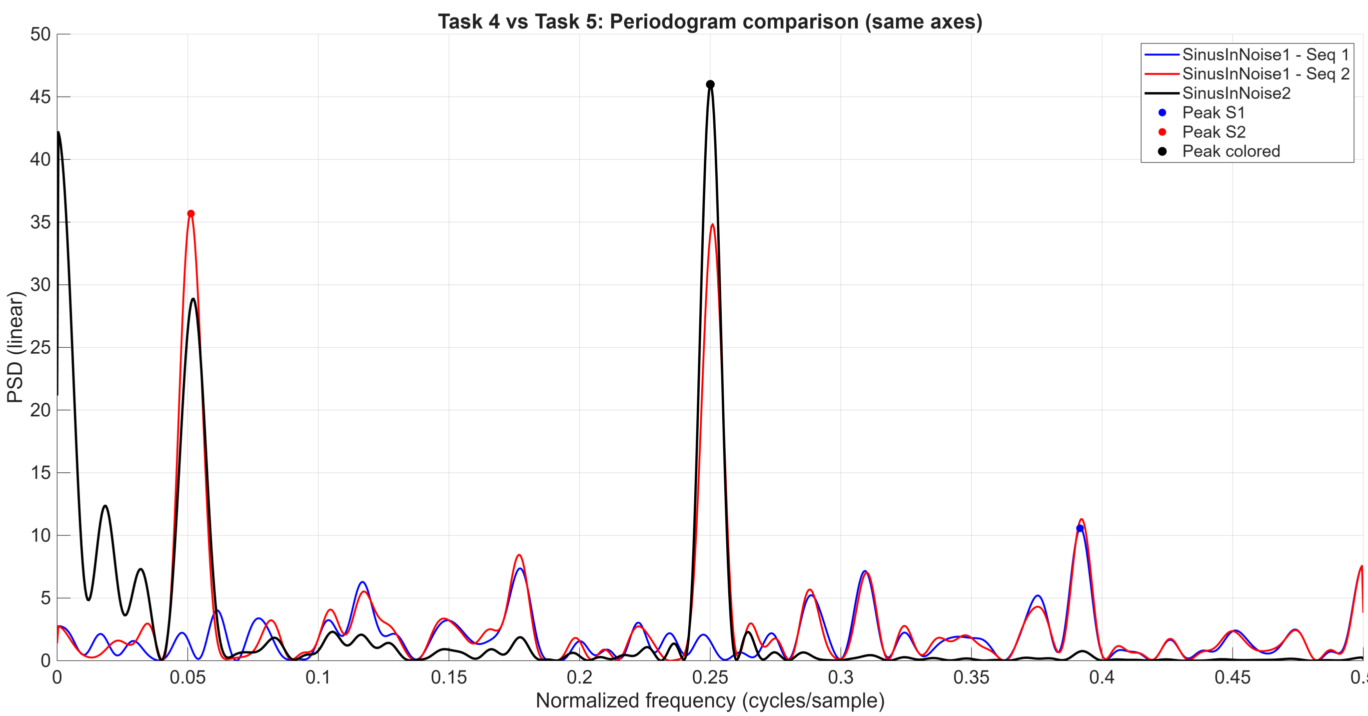


Fig. 4. All three periodograms

VIII. Conclusion

Overall, the study tied together statistical estimation and simple linear–systems analysis under Gaussian models. In the univariate task, the ECDFs and sample statistics from short to long records demonstrated the expected convergence toward the reference $\mathcal{N}(0.5, 2)$ law. In two dimensions, using the general bivariate-normal form with arbitrary means and equal variances clarified how correlation controls ellipse orientation and eccentricity. The datasets were correctly matched to $\rho \approx 0.25$ and $\rho \approx 0.75$. From this

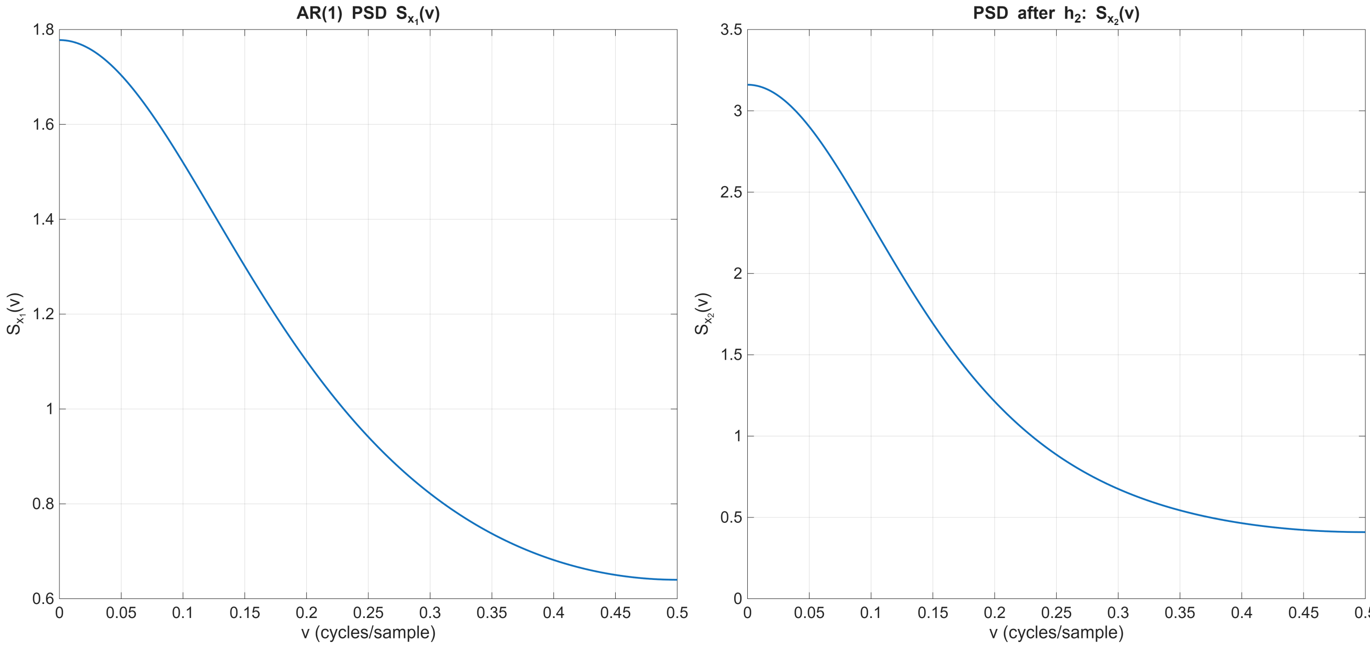


Fig. 5. Task 6: analytic PSDs of $x_1(n)$ and $x_2(n)$ for $\alpha = \beta = 0.25$, $\sigma_z^2 = 1$.

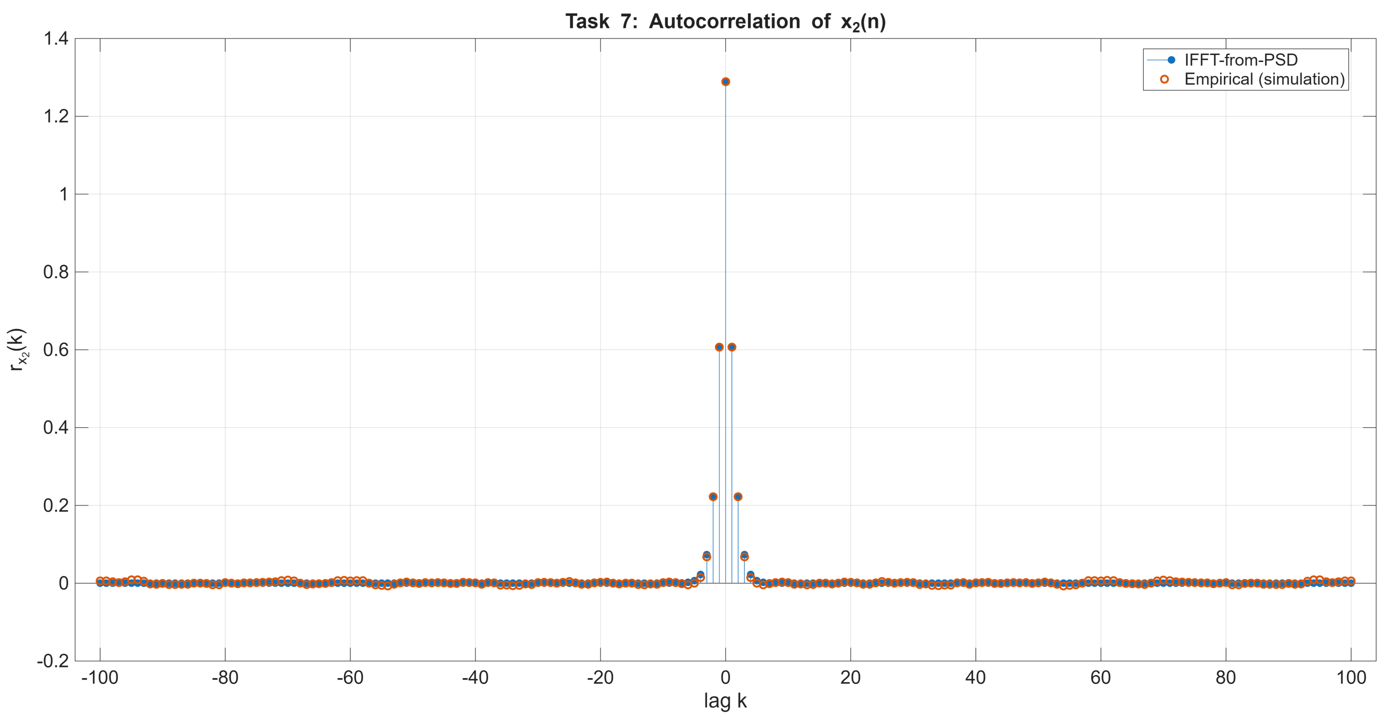


Fig. 6. Task 7: Autocorrelation $r_{x_2}(k)$ from (18) overlaid with IFFT-based and empirical estimates.

joint model, the conditional distribution $X \mid Y = y$ and the sum/difference laws were derived in closed form. In spectral experiments, periodograms cleanly revealed tones for the white-noise case, while under colored noise the low-frequency floor masked the weaker tone and reduced peak salience—degrading the practical accuracy of the recovered frequencies. Finally, for the AR(1) stage followed by a stable one-pole filter, the analytic PSDs $S_{x_1}(\nu)$ and $S_{x_2}(\nu)$ and the closed-form autocorrelation $r_{x_2}(k)$ were obtained. Both the IFFT-based and empirical estimates aligned with these expressions, confirming the theory and illustrating how pole locations set spectral emphasis and correlation length.